

Constraint Interaction and Propagation Regimes in Networked Energy Systems: An Application of Field-Based Dynamics

Abstract

Traditional approaches to energy system stability emphasize load, price, and constraint magnitude as primary drivers of system behavior.

This paper shows that this assumption is incomplete. Using a field-based constraint framework (Castle, 2026a, 2026b, 2026c) applied to a networked model of the Western Interconnect, we demonstrate that system instability is governed not by individual constraint magnitude, but by the interaction structure of constraints across the network.

Across controlled experiments, propagation exhibits invariant hop-ordered structure independent of magnitude. However, when multiple constraints activate simultaneously, the system undergoes discrete regime transitions—from multi-node accumulation to cascading isolation and ultimately network fragmentation—without any change to the underlying propagation mechanism.

These results indicate that instability emerges from relational constraint interactions, not scalar constraint intensity, implying that conventional metrics may fail to detect critical system transitions.

The findings provide a structural basis for identifying instability regimes in networked infrastructure systems and suggest a shift from magnitude-based monitoring to interaction-based analysis.

1. Introduction

Most models of energy system behavior treat constraint magnitude—expressed through load, congestion, or price—as the primary determinant of system stability.

This paper challenges that assumption. We show that system-level instability is not driven by how large individual constraints become, but by how constraints interact across a network. These interactions produce discrete regime transitions that are not explained by magnitude-based metrics alone.

This paper applies a previously developed framework combining signal imbalance, field-based constraint dynamics, and network propagation structure to a modeled electricity system. The objective is not to introduce new theoretical structure, but to evaluate how the existing framework expresses itself under controlled constraint conditions.

The analysis is conducted using a structured representation of the Western Interconnect. Node-level signals are constructed from a combination of synthetic, spatially correlated demand and price

processes designed to reflect realistic system behavior, while the network topology and corridor capacities are grounded in publicly available data sources, including the U.S. Energy Information Administration (EIA, 2024), CAISO market data interfaces (CAISO, 2024), and published WECC transmission maps (WECC, 2023). Data adapters for live EIA and CAISO feeds are incorporated into the modeling framework, enabling extension of the analysis to real-time data.

We focus specifically on how increasing constraint interaction alters propagation behavior. By systematically varying loading and constraint activation across a networked system, we identify distinct propagation regimes and characterize the conditions under which they emerge.

2. Framework

We briefly summarize the applied framework; full derivations are provided in prior work.

System state is represented by two normalized observables:

- (S) : local pressure (e.g., demand)
- (R) : constraint expression (e.g., price)

The difference defines a local imbalance:

$$\Phi = R - S$$

Propagation is modeled as a constrained field evolving over a network:

- Nodes represent system regions
- Edges represent transmission corridors with finite capacity
- Edge constraints limit admissible differences $|\Delta \Phi_{ij}|$

Transmission occurs through local interactions, with effective propagation determined by both network topology and constraint activation.

No additional mechanisms are introduced in this application.

3. Experimental Design

3.1 Network Structure

The system is represented as a 14-node graph corresponding to major balancing authorities in the Western Interconnect, with 28 transmission corridors.

Two primary structural regions are identified:

- Northwest spine (high-capacity corridors)
- Southwest spine (lower-capacity, bottleneck-prone corridors)

Corridor Loading Experiment: NW_spine ($\eta=0.2$ $\gamma=0.02$ steps=60)

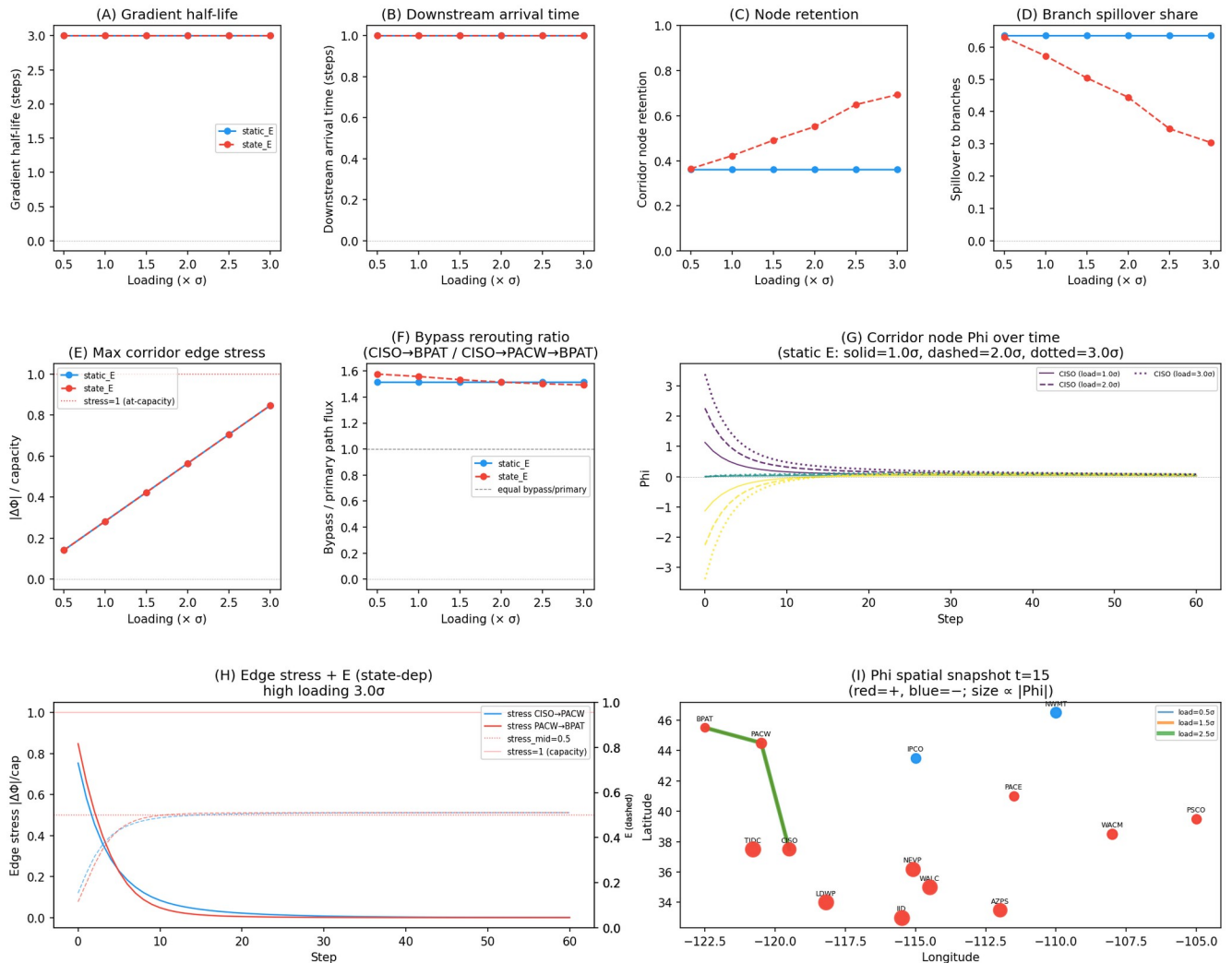


Figure 1: CL_NW_spine.png — Northwest transmission spine

Corridor Loading Experiment: SW_spine ($\eta=0.2$ $\gamma=0.02$ steps=60)

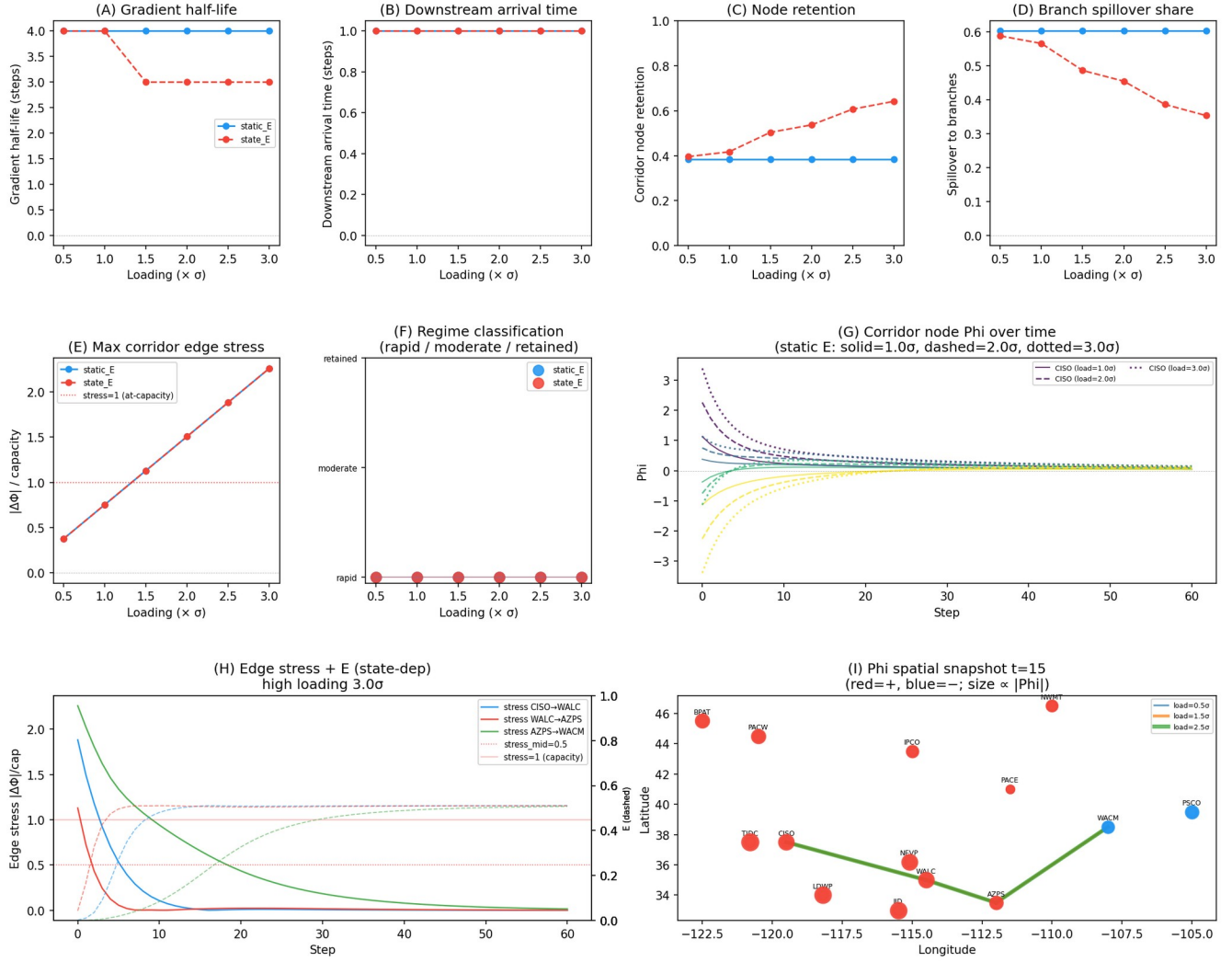


Figure 2: CL_SW_spine.png — Southwest transmission spine

3.2 Variables and Construction

For each node:

- (S): normalized demand signal
- (R): normalized price signal
- ($\Phi = R - S$)

Edge constraints are defined by capacity-normalized thresholds:

$$|\Delta \Phi_{ij}| \leq \theta_{ij}$$

Thresholds are expressed relative to background variability (σ).

3.3 Experimental Sequence

The following experiments are conducted:

1. Propagation structure (impulse response)
2. Mediation patterns
3. Threshold behavior
4. Corridor loading
5. Hard constraint activation
6. Multi-constraint cascade (single, dual, triple constraints)

4. Observations

4.1 Propagation Structure

Propagation exhibits consistent hop-ordered behavior across all tested conditions. Ordering remains stable across thresholds ranging from 1.0σ to 2.5σ , indicating that propagation structure is determined by network topology rather than magnitude activation.

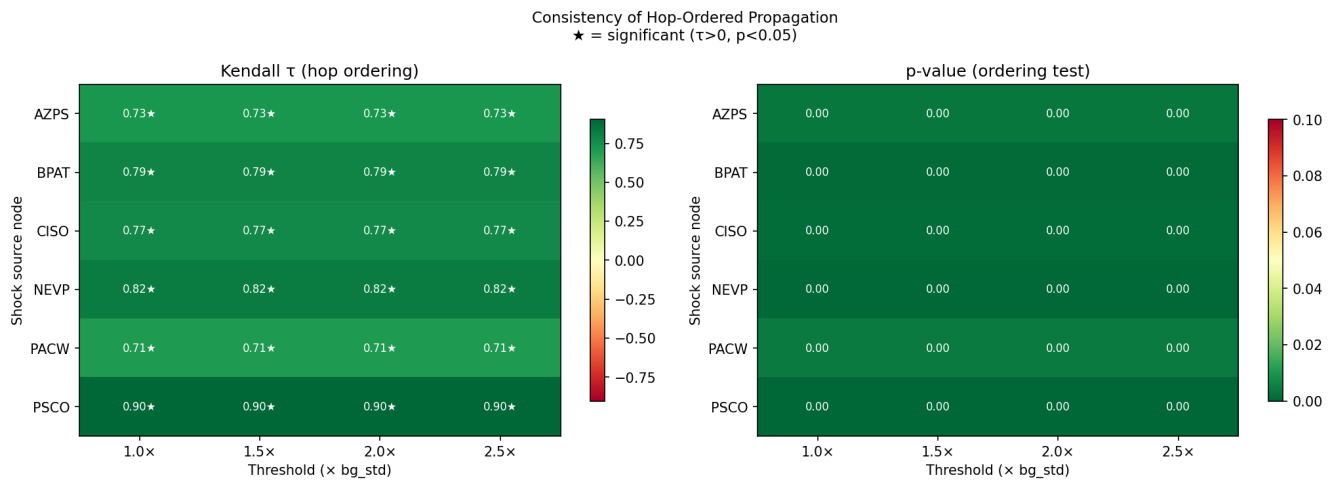


Figure 3: T1_consistency_matrix.png — Propagation consistency matrix

4.2 Mediation Structure

Transmission is mediated by identifiable but non-universal nodes and corridors. The PACW–BPAT corridor and WALC node appear frequently as intermediaries, reflecting their structural position within the network. However, no node or corridor mediates transmission across all conditions.

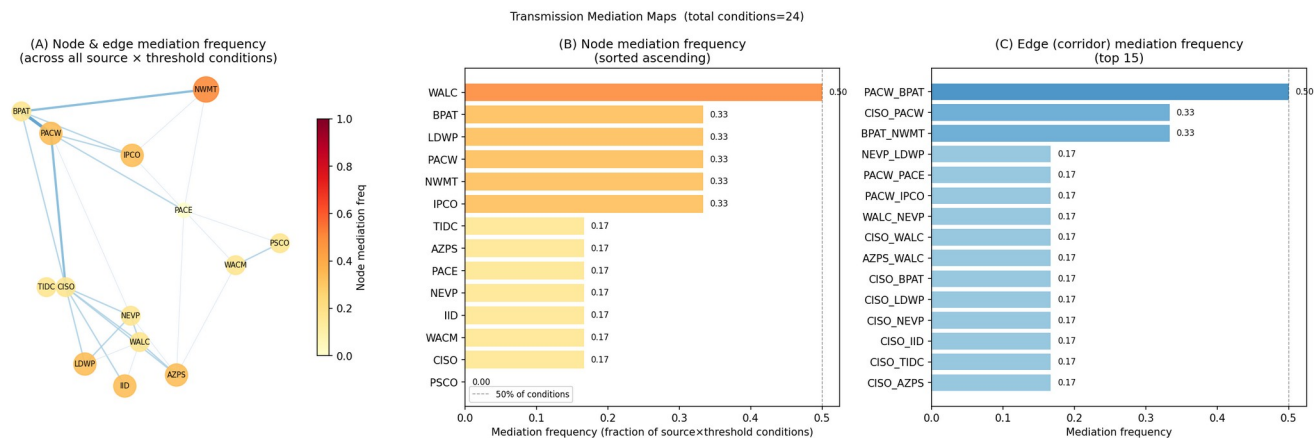


Figure 4: T2_mediation_maps.png — Mediation frequency map

4.3 Threshold Behavior

Thresholds influence the magnitude of propagation but do not determine whether propagation occurs. Ordering remains invariant across thresholds, indicating that constraint activation affects intensity rather than structural propagation.

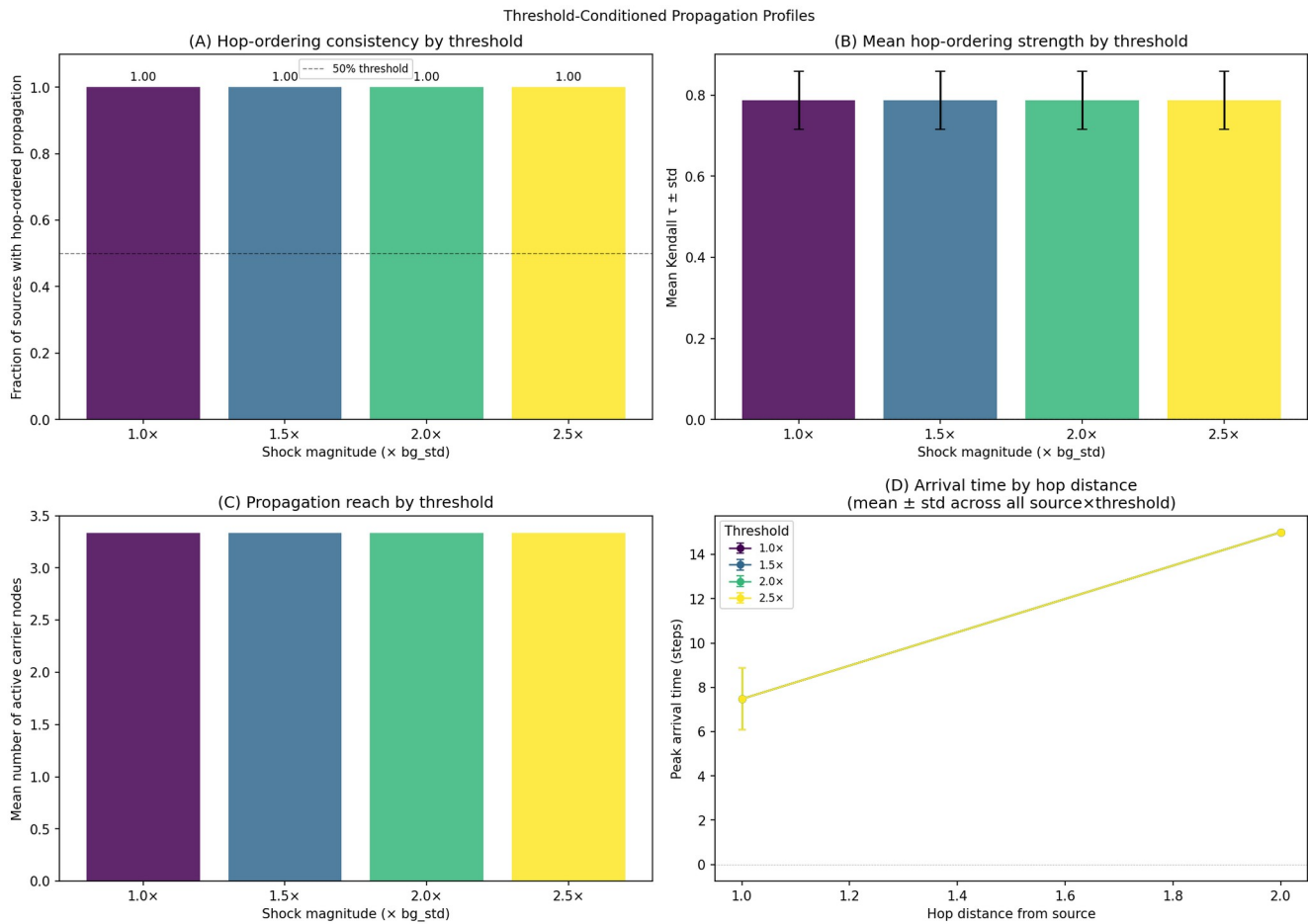


Figure 5: *T3_threshold_profiles.png* — Threshold activation profiles

4.4 Source Dependence

Propagation patterns vary with source location. Different nodes engage distinct transmission paths, and no universal propagation corridor exists.

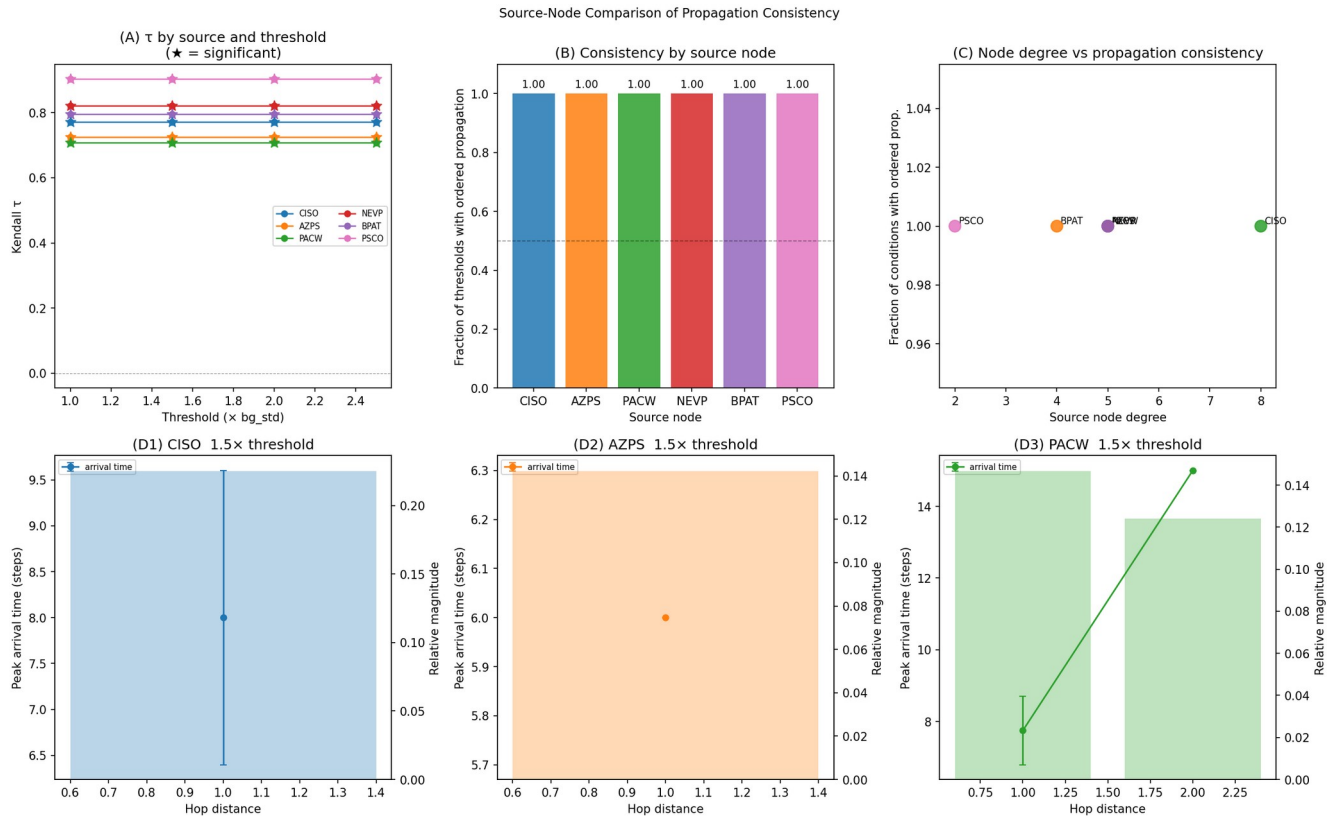


Figure 6: *T4_source_comparison.png* — Source comparison

4.5 Corridor Loading (Single Constraint)

Increasing loading on individual corridors produces greater local accumulation and reduced spillover, but does not alter overall propagation structure. The system remains connected and continues to disperse imbalance.

Corridor Loading Experiment: NW_spine ($\eta=0.2$ $\gamma=0.02$ steps=60)

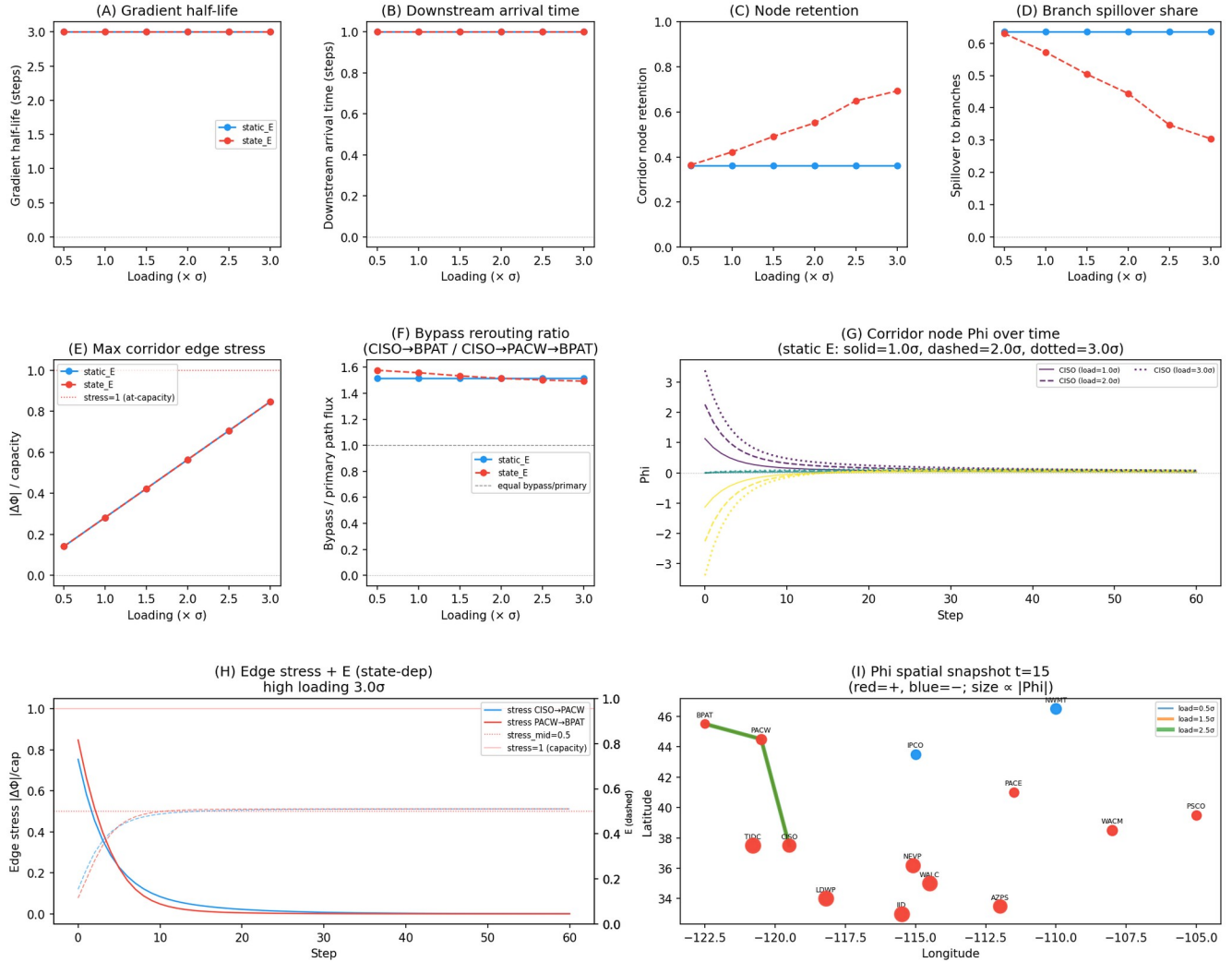


Figure 7: CL_NW_spine.png — NW loading behavior

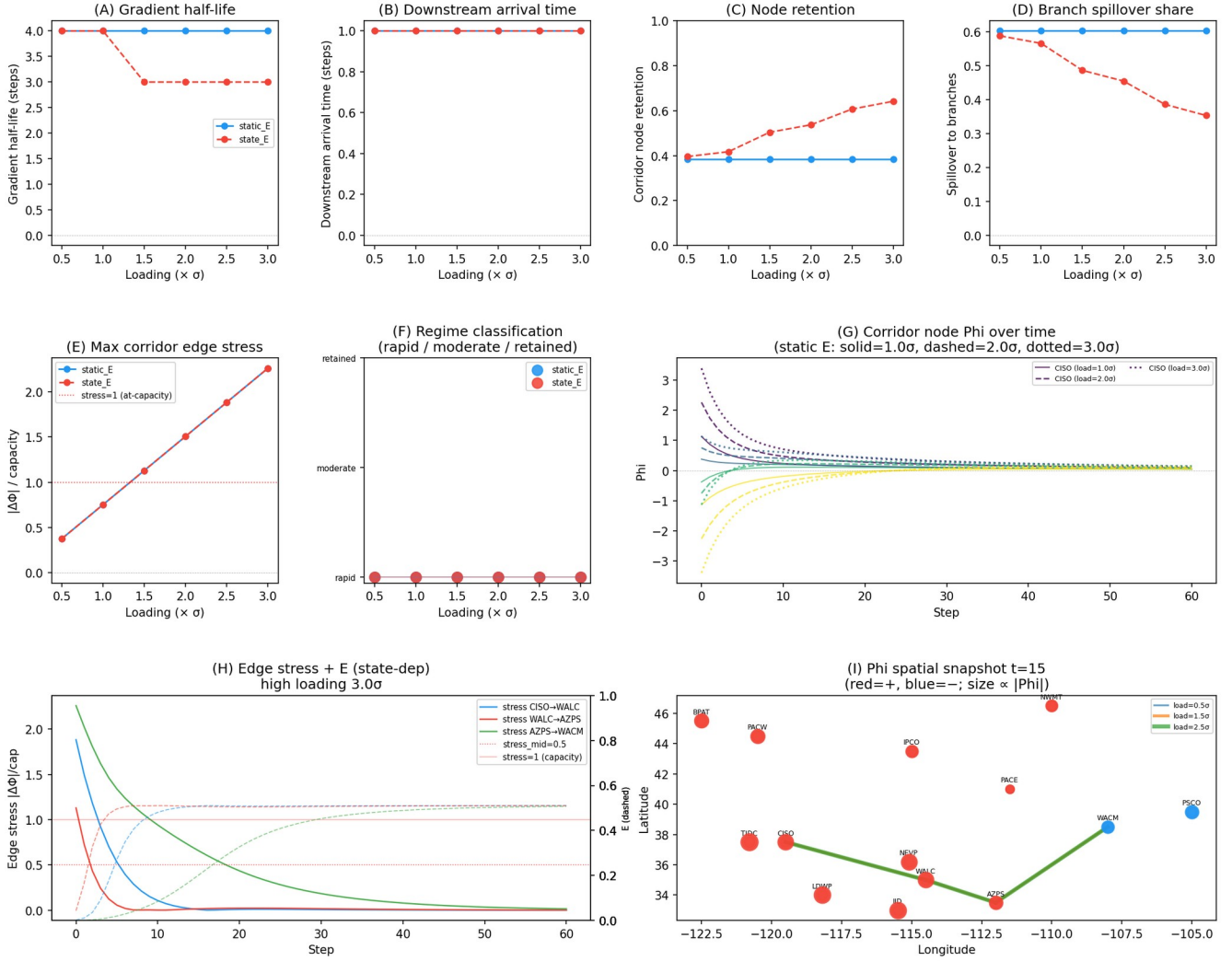


Figure 8: CL_SW_spine.png — SW loading behavior

4.6 Hard Constraint Activation

Introducing hard constraints increases local retention of imbalance and reduces transmissibility. However, even under strong constraint activation, the system does not exhibit structural fragmentation when only a single constraint is active.

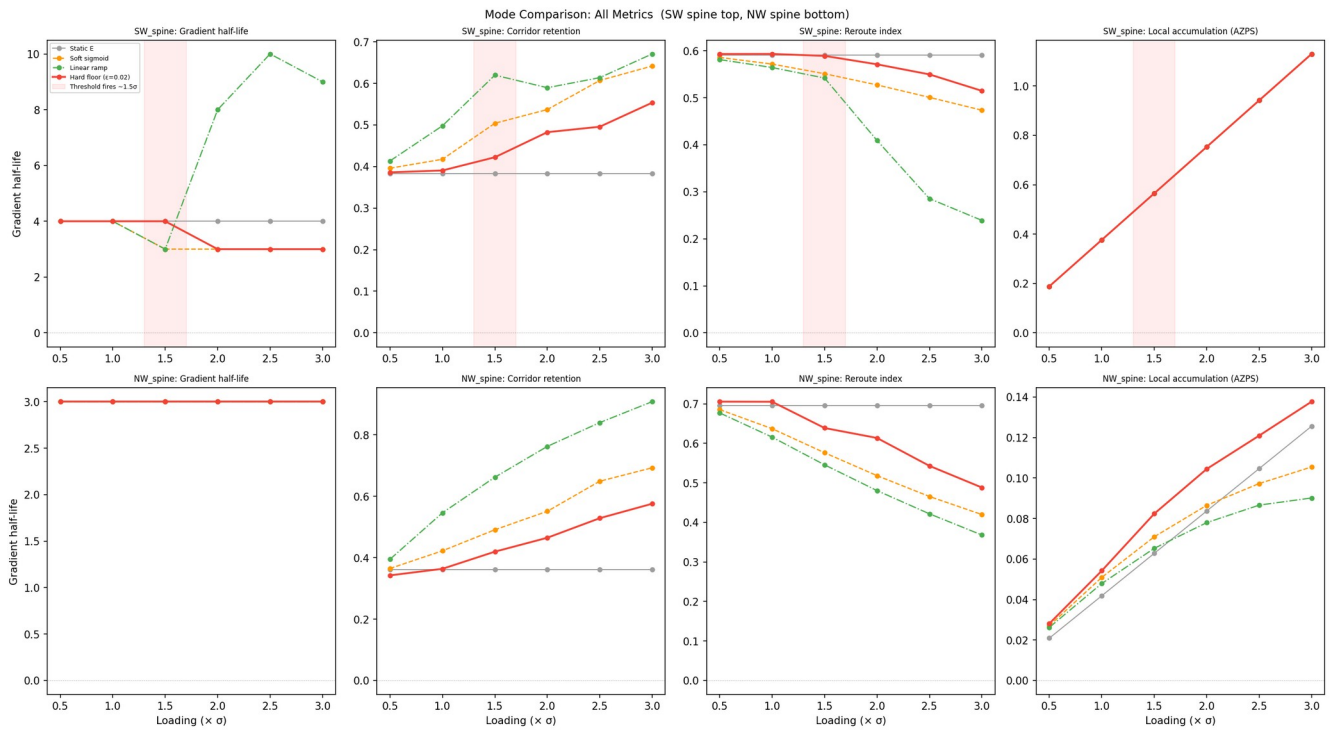


Figure 9: HF_mode_comparison.png — Mode comparison

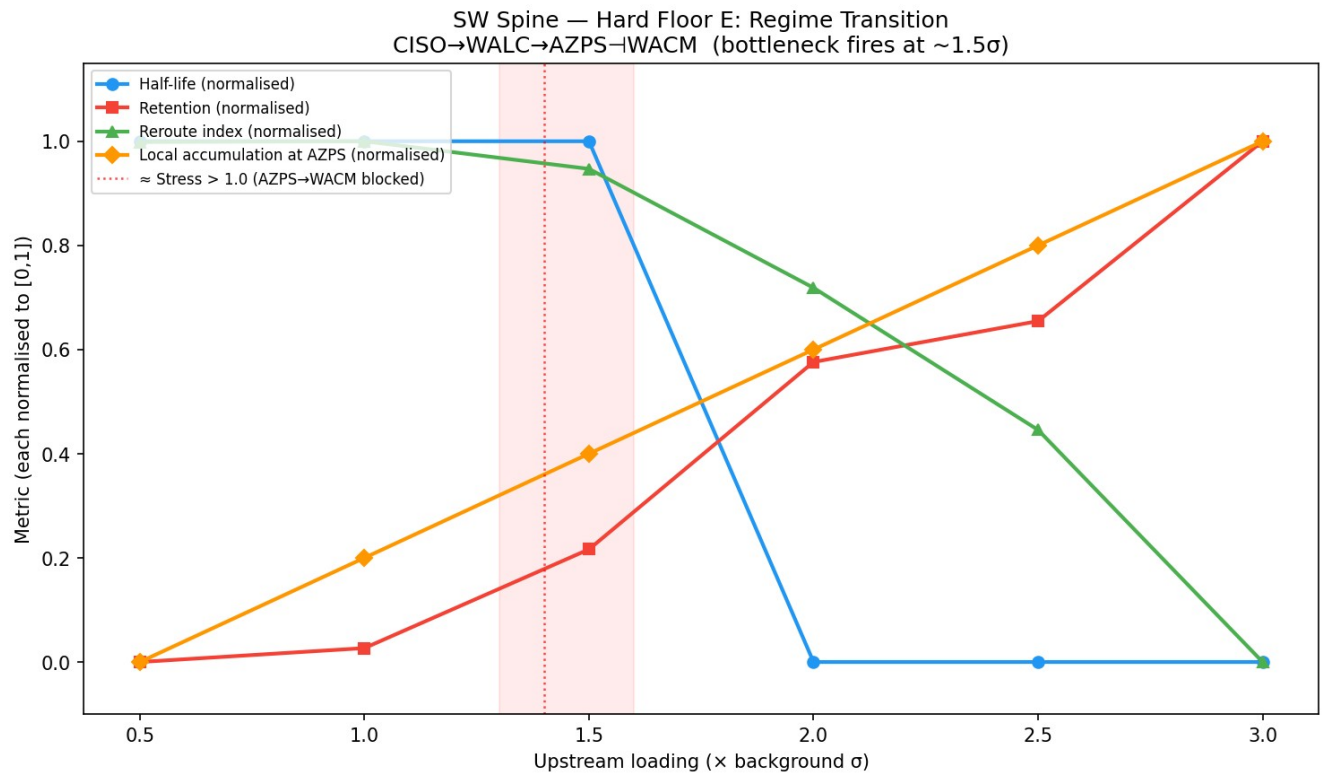


Figure 10: HF_regime_transition.png — Hard constraint transition

4.7 Variant Behavior

Across variations in damping and constraint formulation, the system exhibits increased retention and localized effects but does not produce structural isolation or fragmentation under single-constraint conditions.

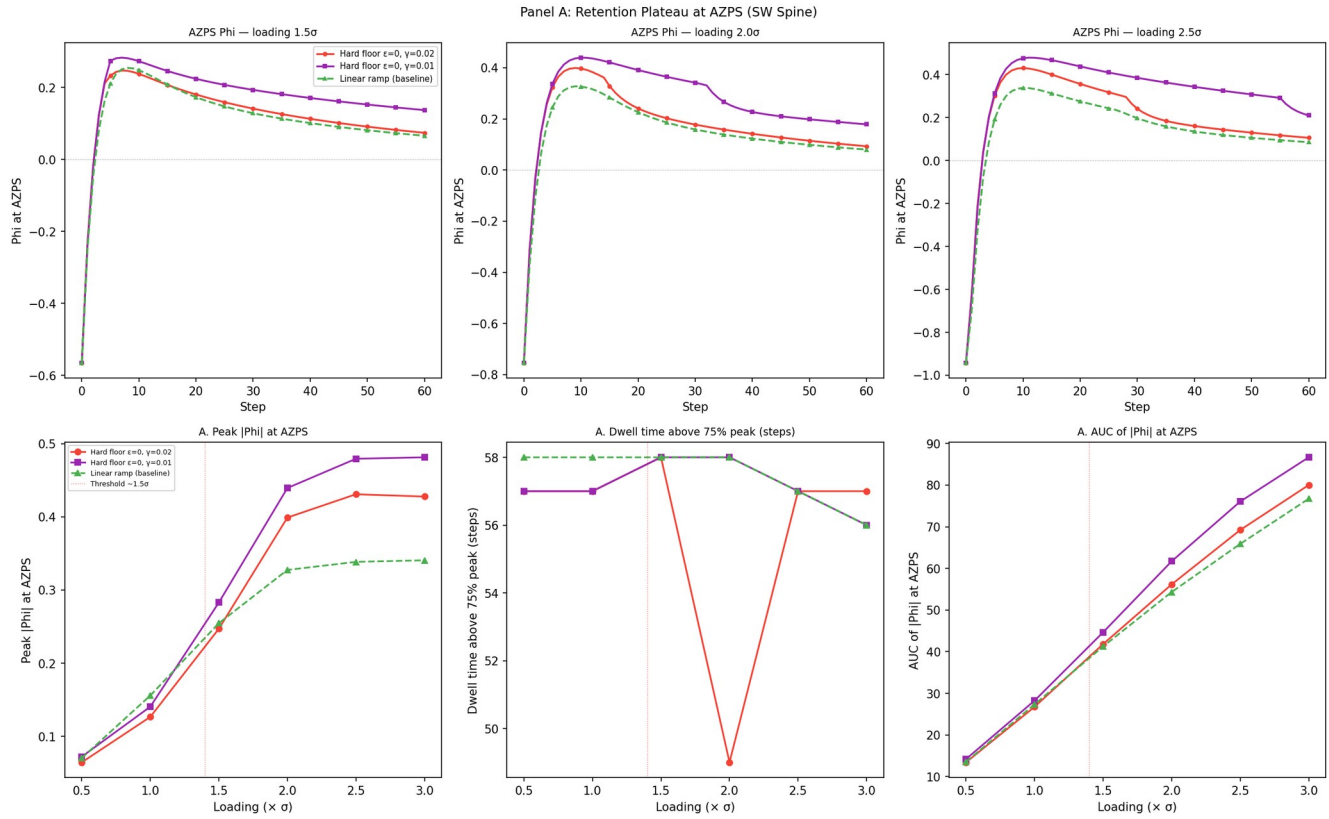


Figure 11: SV_A_retention.png — Retention

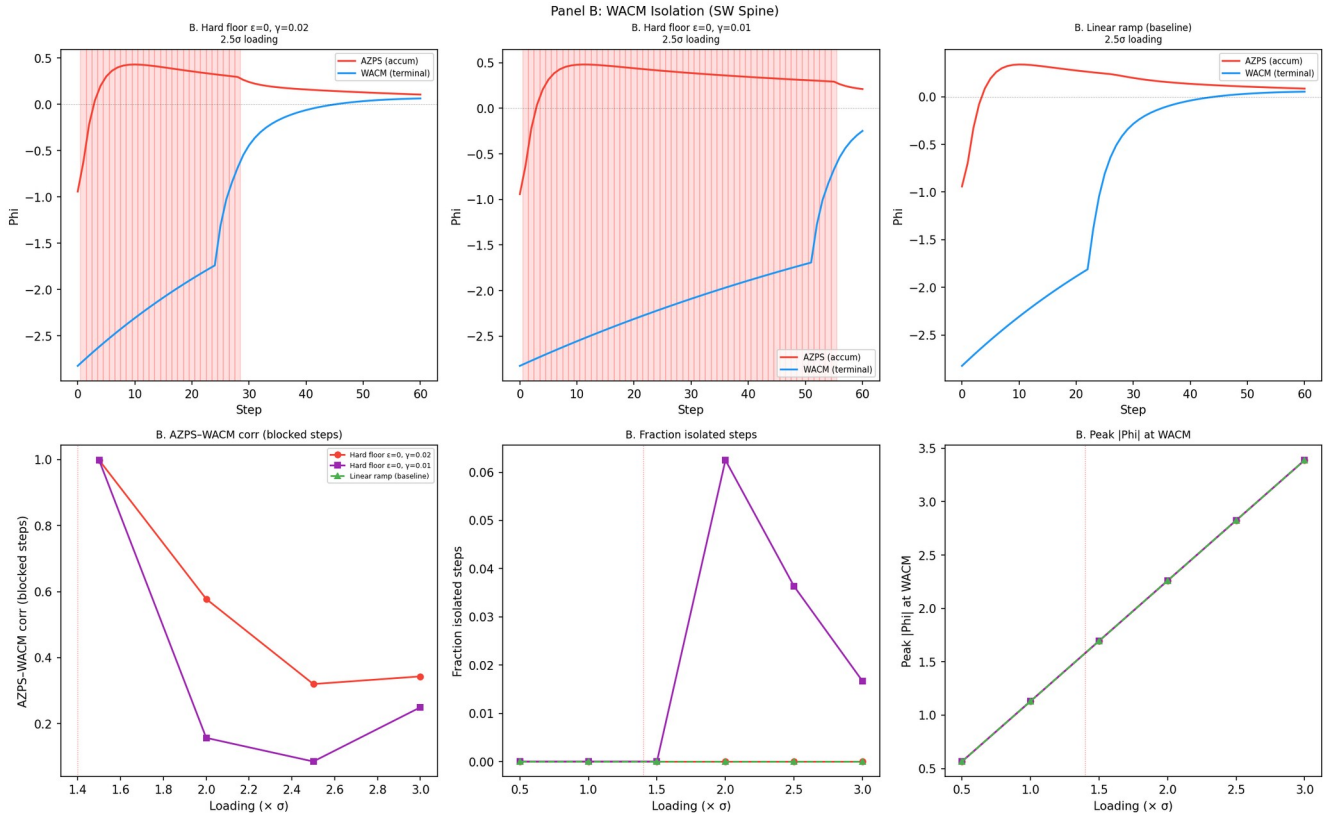


Figure 12: SV_B_isolation.png — Isolation

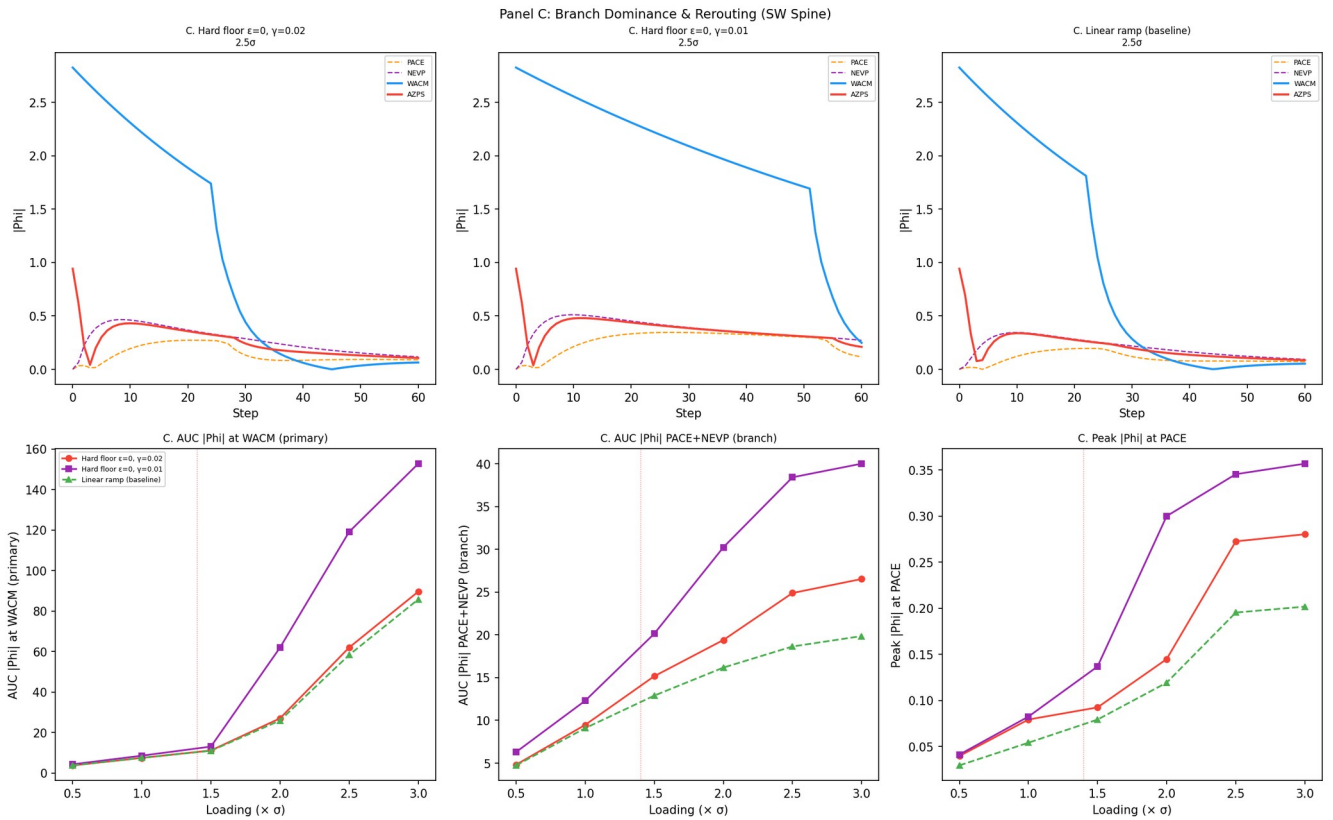


Figure 13: Figure 14: SV_C_branches.png — Branching

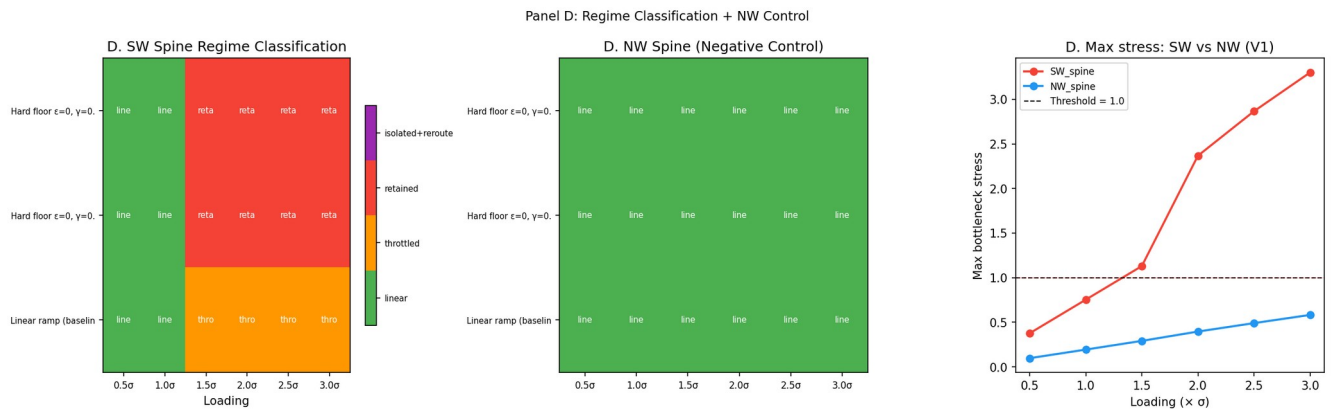


Figure 14: *SV_D_regime.png* — Regime classification

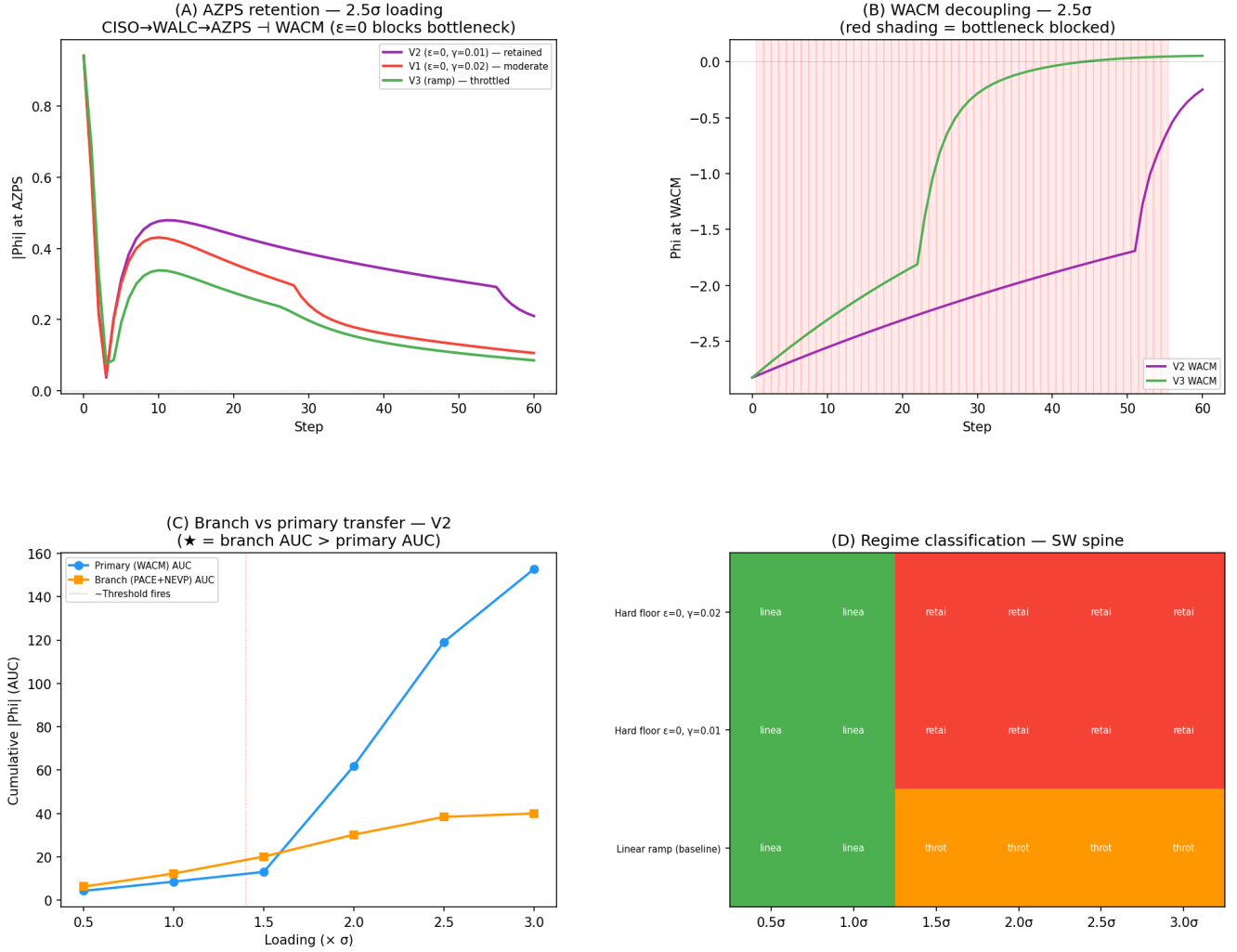


Figure 15: SV_summary.png — Variant summary

4.7.1 Elasticity As A Routing-Cost Field

While aggregate propagation behavior remained structurally consistent across elasticity variants, additional experiments indicate that elasticity primarily alters *routing behavior* rather than aggregate prediction error.

Propagation may be expressed as a constrained field evolution:

$$\Phi_{t+1} = L(E, \theta) \Phi_t$$

where:

- Φ_t represents the network imbalance field,
- $L(E, \theta)$ represents a state-dependent propagation operator,
- E modulates effective transmissibility,

- and θ represents corridor constraint thresholds.

Under this formulation, elasticity acts as a routing-cost field governing the admissibility of stress propagation across network corridors.

To evaluate this effect, cross-corridor leakage experiments were conducted under multiple elasticity regimes, including:

- uniform transmissibility,
- calibrated static elasticity,
- state-dependent elasticity,
- shuffled elasticity,
- and inverted elasticity configurations.

Aggregate RMSE differed minimally across regimes, indicating that elasticity contributes little to overall prediction error. However, cross-corridor leakage behavior changed substantially.

At symmetric loading conditions (2.5σ equivalent), calibrated elasticity reduced SW to NW leakage from 6.25 (uniform) to 3.60, while state-dependent elasticity reduced leakage further to 1.44. Moran's I remained approximately invariant across regimes, indicating that elasticity alters routing allocation rather than large-scale spatial autocorrelation structure.

These findings indicate that elasticity primarily governs:

- stress retention,
- corridor transmissibility,
- and cross-network routing behavior,

rather than aggregate field magnitude.

State-dependent elasticity produced the strongest suppression under high loading conditions, causing imbalance to accumulate locally rather than propagate across connector corridors. This behavior is consistent with the observed accumulation and isolation regimes described in subsequent cascade experiments.

Shuffled and inverted elasticity configurations produced little change relative to calibrated static elasticity. Analysis showed that rolling normalization removed most corridor-level mean differences in the synthetic field, leaving similar mean elasticity values across major corridors. As a result, temporal correlation structure alone was insufficient to reverse routing preference under synthetic conditions.

These results support interpreting elasticity as a routing-cost field rather than a prediction-error correction term. Under constrained loading, lower elasticity increases effective propagation resistance, modifies admissible transmission pathways, and suppresses inter-corridor stress transfer.

Figure 6: Routing Outcomes — Real vs Shuffled vs Inverted vs Uniform E

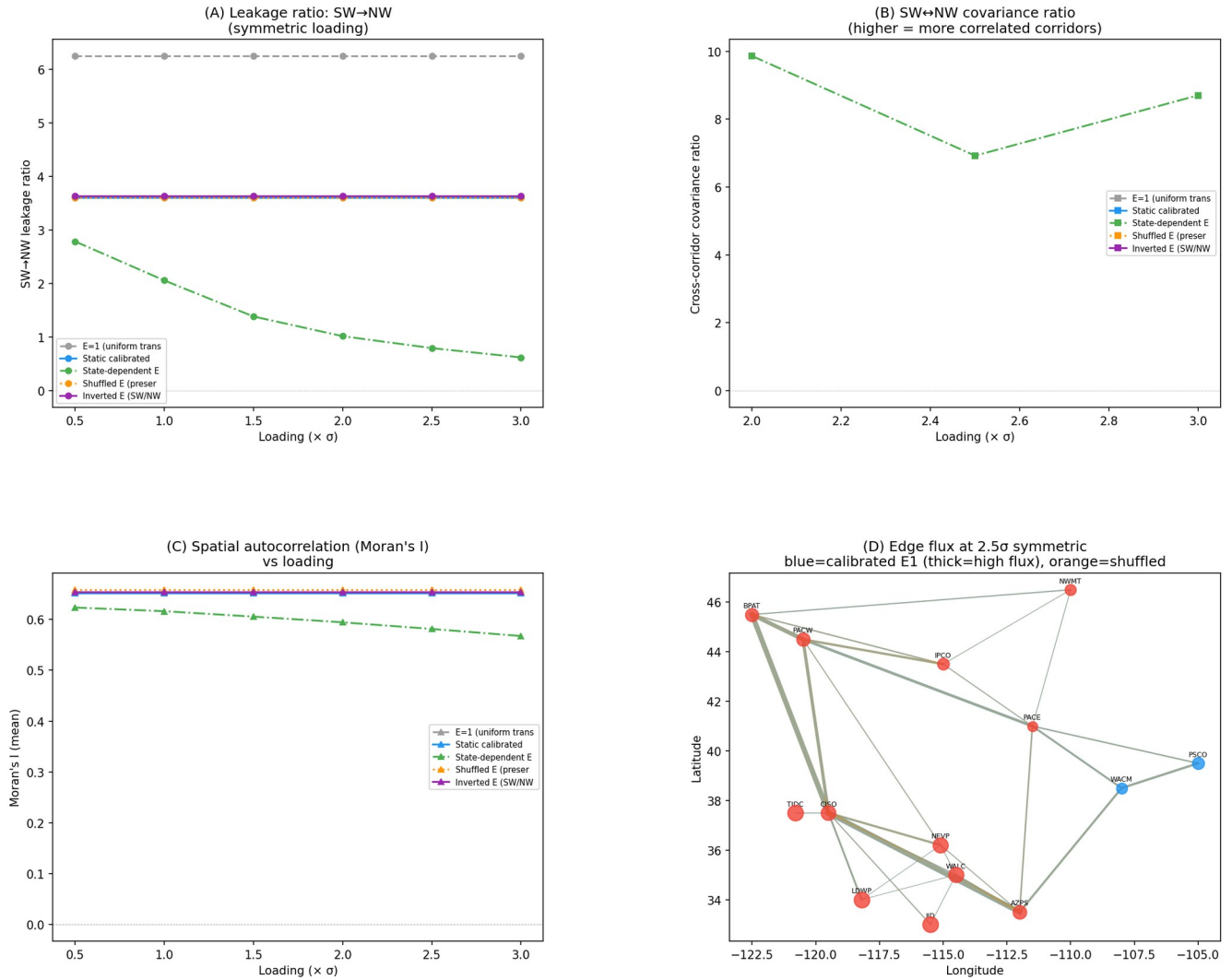


Figure 16: CC_fig6_routing_comparison.png — Cross-Corridor Experiment Figure 6. Routing Outcomes - Real vs Shuffles vs Inverted vs Uniform E

4.8 Multi-Constraint Cascade

When multiple constraints are activated simultaneously, system behavior changes qualitatively.

4.8.1 Overview

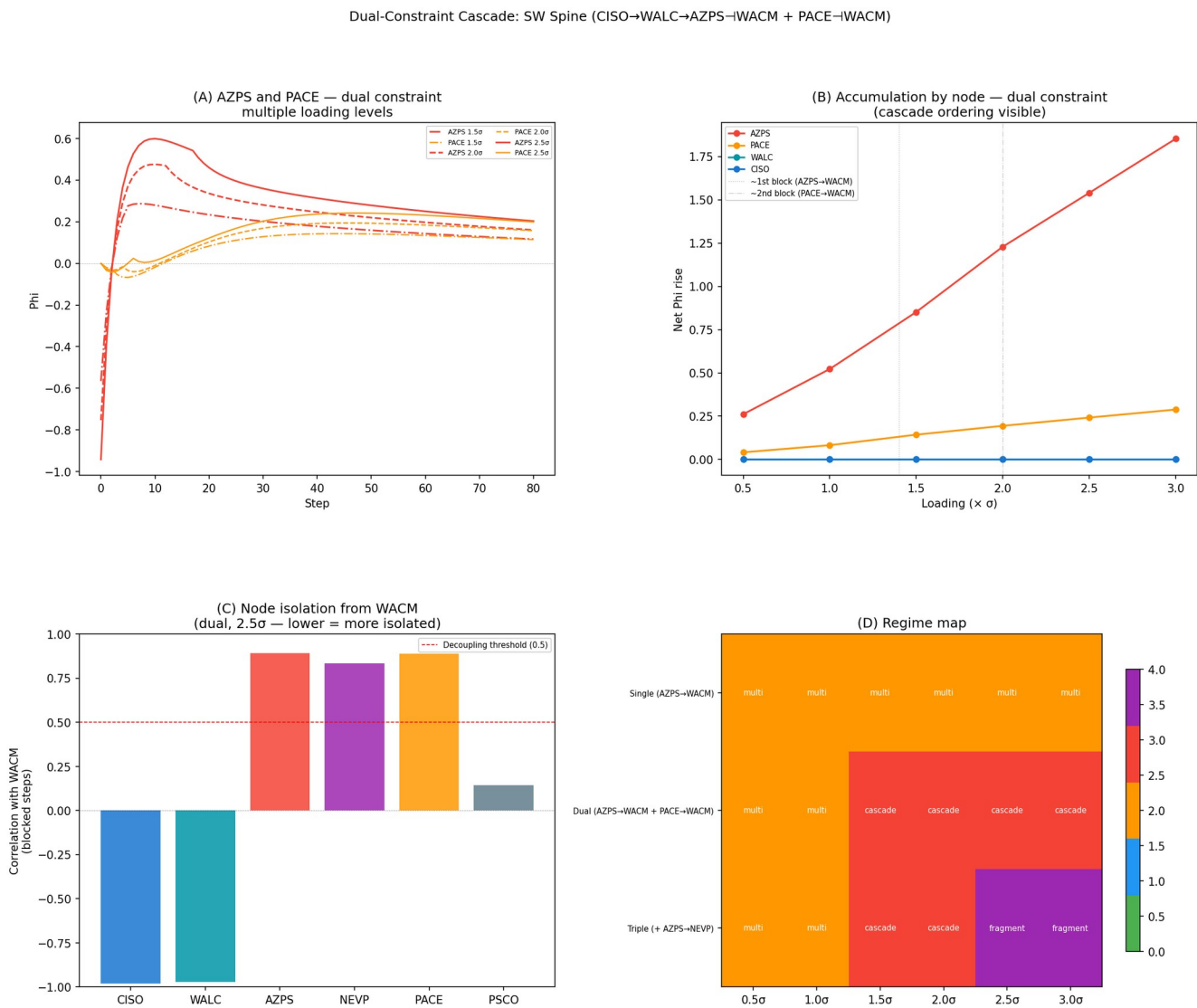


Figure 17: DC_dashboard.png — Cascade overview

4.8.2 Accumulation

Dual constraints produce multi-node accumulation, with imbalance distributed across upstream nodes rather than resolving at constrained sinks.

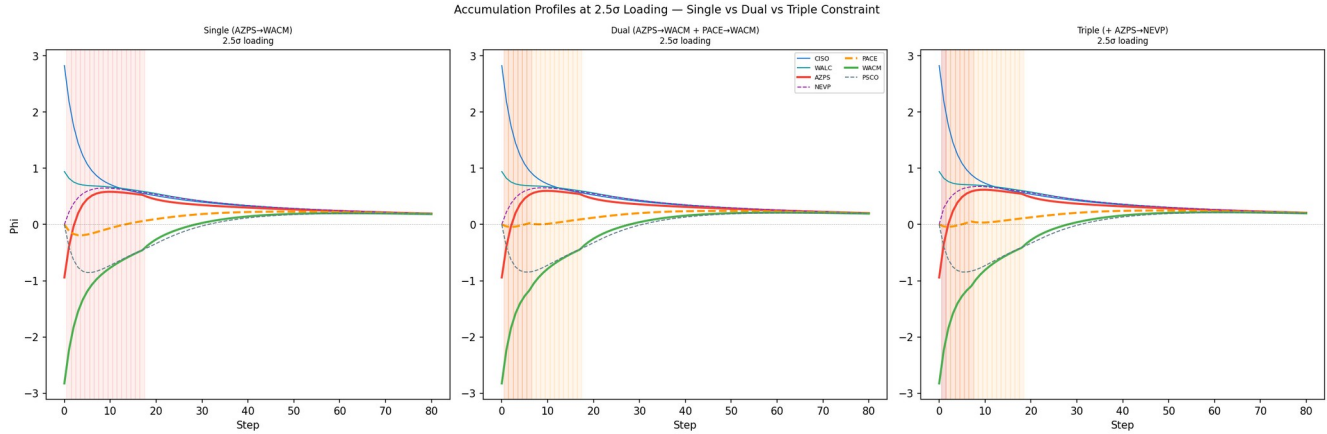


Figure 18: DC_accumulation.png — Accumulation patterns

4.8.3 Timing

Constraint activation occurs within a shared temporal window. Primary constraints persist longer than secondary constraints, creating a period during which imbalance cannot be resolved downstream.

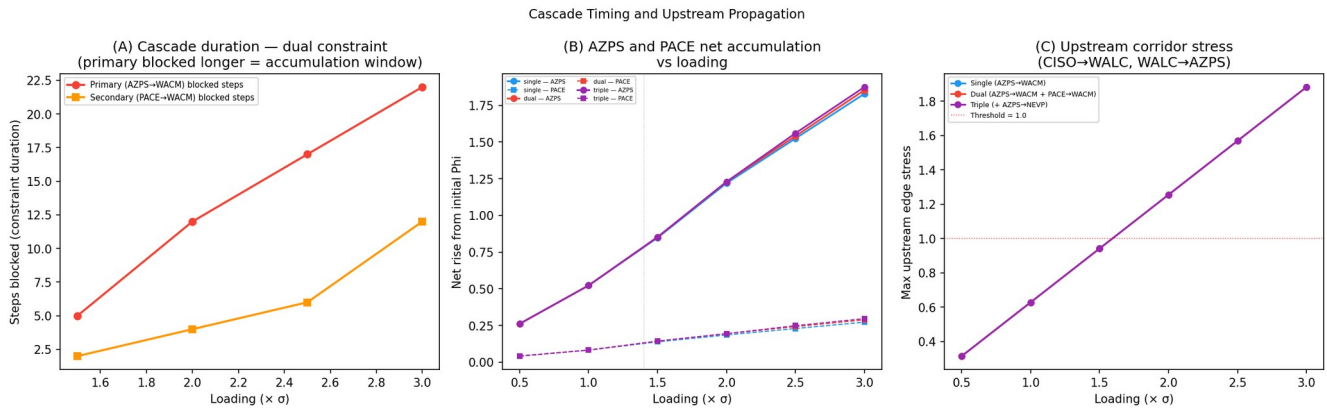


Figure 19: DC_cascade_timing.png — Cascade timing

4.8.4 Isolation

Intermediate nodes remain inactive, and downstream nodes become effectively decoupled from upstream propagation.

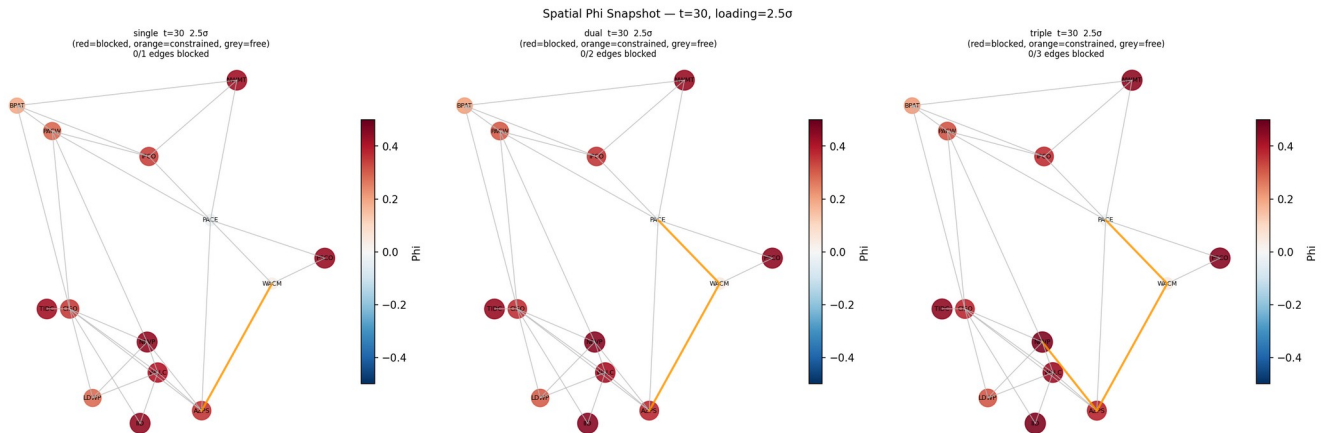


Figure 20: DC_isolation_map.png — Isolation map

4.8.5 Regime Transition

System behavior transitions across three regimes:

- Single constraint: multi-node accumulation
- Dual constraint: cascading isolation
- Triple constraint: network fragmentation

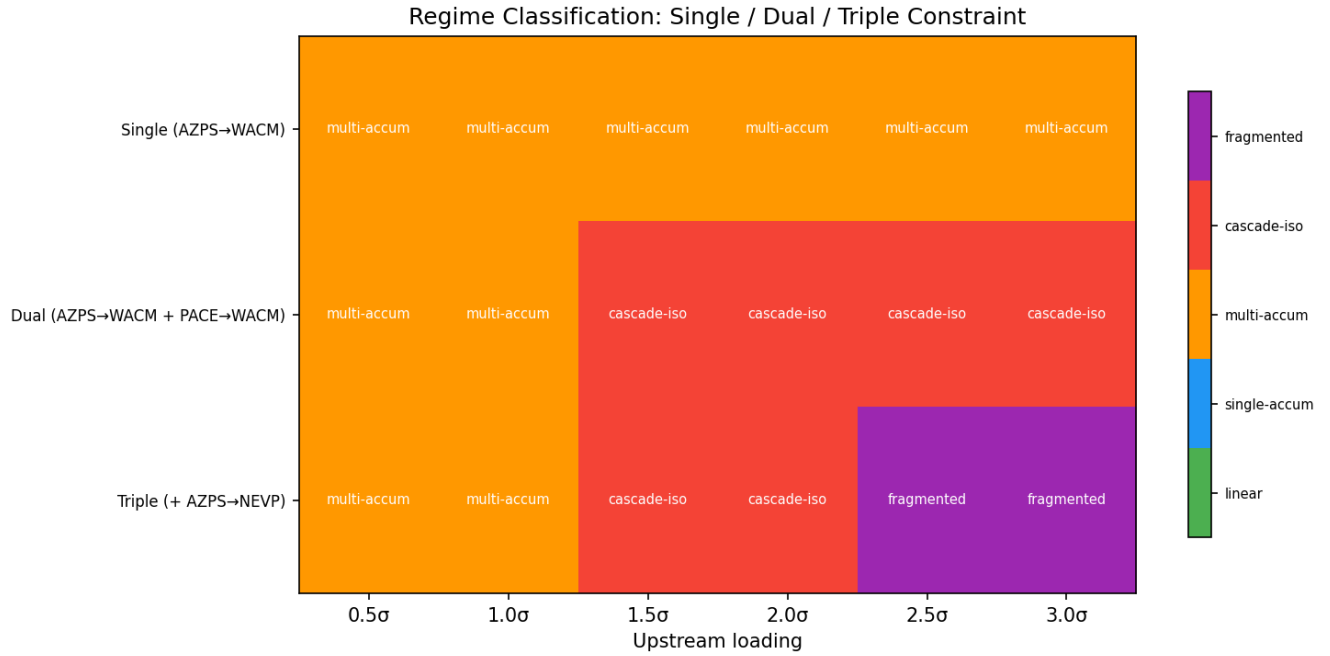


Figure 21: *DC_regime_map.png* — Regime map

5. Interpretation

These behaviors are consistent with the constraint-field formulation (Castle, 2026b) and network interaction structure (Castle, 2026c).

- From a signal perspective, constraint activation limits expression, resulting in accumulation of imbalance.
- From a field perspective, interacting constraints introduce discontinuities, preventing smooth propagation.
- From a network perspective, constraint activation removes effective transmission paths, producing state-dependent topology.

These effects arise without modification to the underlying propagation mechanism.

This suggests early-warning indicators should track multi-constraint activation patterns rather than aggregate load or price levels.

6. Key Result

Constraint interaction, rather than individual constraint magnitude, governs the transition from continuous propagation to isolated and fragmented system behavior.

A magnitude-based interpretation of system stress would predict gradual degradation under increasing load. Instead, we observe discrete regime transitions driven by constraint interaction, indicating that conventional monitoring approaches may systematically mischaracterize system risk.

7. Limitations

- Synthetic signal construction
- Parameter sensitivity
- Network approximation

8. Conclusion

This application demonstrates that the field-based constraint framework produces distinct propagation regimes under interacting constraints. While single constraints produce localized effects, interacting constraints induce qualitative changes in system behavior, including isolation and fragmentation. These results highlight the importance of constraint interaction in determining system-level outcomes.

9. Additional Information

All simulation procedures and data construction methods are described in the manuscript. Implementation details and code are available from the author upon reasonable request.

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